

The empirical recurrence for OEIS A302962, recovered from its tail

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Abstract

OEIS A302962 records “Empirical recurrence of order 80” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 129$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 82$. Its last coefficient is $c_{80} = -2097152 \neq 0$, so no further tail satisfies anything shorter; any order-80 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A302962 is “Number of nX5 0..1 arrays with every element equal to 0, 1, 2, 3 or 6 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

16, 384, 3924, 43588, 493132, 5602382, 63612987, 722646394, ...

The entry states:

Empirical recurrence of order 80 (see link above)

As of the “Last modified” line on the live entry (Apr 16 2018 11:53:18, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 129$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 129$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 80: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 80.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 258$ exact terms from $n = 2$ on, it returns order 80 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{80} c_i a(n-i),$$

c_1	18	c_{21}	29976563273	c_{41}	−49236809166104	c_{61}	317570270659916
c_2	−63	c_{22}	18946703594	c_{42}	−1051659012681	c_{62}	−243619661342912
c_3	−133	c_{23}	36643458785	c_{43}	180244888989111	c_{63}	173176986675324
c_4	−782	c_{24}	−174252161925	c_{44}	−53472164811510	c_{64}	−115019795606852
c_5	7041	c_{25}	−7292426848	c_{45}	70825362279851	c_{65}	75021810083296
c_6	18527	c_{26}	−59529366224	c_{46}	−371359813979235	c_{66}	−46635443591392
c_7	−99336	c_{27}	670065725417	c_{47}	332686374049502	c_{67}	26873544190760
c_8	−52083	c_{28}	−308526003436	c_{48}	−306926386184583	c_{68}	−14646280759136
c_9	−146819	c_{29}	173161241222	c_{49}	624087396333042	c_{69}	7580925042848
c_{10}	684858	c_{30}	−2148047205813	c_{50}	−685930407990734	c_{70}	−3661277737472
c_{11}	5506135	c_{31}	1420856885278	c_{51}	621270060801942	c_{71}	1623320414688
c_{12}	6719883	c_{32}	−56057700594	c_{52}	−818720998094200	c_{72}	−633759265344
c_{13}	−27873426	c_{33}	6789290241916	c_{53}	888883972979692	c_{73}	209087821824
c_{14}	−193534538	c_{34}	−3839567386127	c_{54}	−788152103797366	c_{74}	−57434758528
c_{15}	106484675	c_{35}	−8370289143753	c_{55}	814341108348754	c_{75}	12343690752
c_{16}	690570131	c_{36}	−15311892892657	c_{56}	−810115244294603	c_{76}	−1409021952
c_{17}	1982313999	c_{37}	18759115018924	c_{57}	686136108439594	c_{77}	−317411328
c_{18}	−3328173627	c_{38}	36724611270792	c_{58}	−600414625020440	c_{78}	147292160
c_{19}	−5285311702	c_{39}	11520543795691	c_{59}	530276251671986	c_{79}	−7340032
c_{20}	−10380336854	c_{40}	−69710585392610	c_{60}	−417307506584766	c_{80}	−2097152

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 82$, and it is the recurrence OEIS A302962 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{80} - c_1 x^{79} - \dots - c_{80}$. Its constant term is $-c_{80} = 2097152$, which is not zero, so $q_{\min}(0) \neq 0$ and

$q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 80 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 80, so $\deg q = 80$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 80, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 80, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -2097152 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-80 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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